

Kurnosenko Vitaly Nikolaevich

Antifraud Analyst | Fraud Rule Development | ML-based Detection | Payment Risk | Data Analysis (SQL / Python)

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Languages: Russian — native; English — B2 working proficiency; Chinese (Mandarin) — HSK 4

PROFESSIONAL SUMMARY

Engineer and analyst with **direct hands-on experience in antifraud systems** — fraud rule design, ML-based bot and abuse detection, order validation pipelines, threshold tuning, and back-testing. Worked **3+ years** on the Yofi/Botnot anti-fraud platform (USA) serving enterprise Shopify merchants including Lululemon, building and maintaining the full fraud detection stack in **Python, SQL, and dbt**.

Additional layer: **15+ years** in corporate finance and banking operations, giving deep understanding of payment flows, transaction structures, and financial risk — rare context for an analyst working on payment fraud typologies.

PhD in Physics and Mathematics — strong grounding in probability theory and statistical modeling directly applicable to rule back-testing, false positive analysis, and fraud scoring.

ROLE FIT — ANTIFRAUD ANALYST MIDDLE

JD Requirement	My Evidence
Test and improve antifraud rules	Maintained `yofi-rules-monorepo` — per-shop YAML rule configs, threshold tuning (`min_return_rate`, `min_fraud_tags`), allowlist/blocklist management, pre-commit parity checks between Python and dbt
Optimize rules across channels (card, transfers, tokenization)	Order validation pipeline: BIN validation, IP checks, email, phone, user agent, shipping/billing address signals — multi-signal order risk assessment
Fraud detection automation	ML routing pipeline — `ShopifyClaimAbusersModel`, `ShopifyReturnFraudstersModel`, `ShopifyResellersModel`, `FTID_FRAUD_SCORE`, `BOT_ABUSE_SCORE`; SageMaker-backed real-time scoring via Lambda
Back-testing and efficiency assessment	Feature analytics Lambda — return rate calculations, refund line items analysis, pattern detection in historical order data persisted to BigQuery/Spanner
Analyze confirmed fraud cases, identify new patterns	`is_bad_actor`, `is_blacklisted`, trust/risk signal system; return analytics processor for new fraud typologies
Monitor metrics, build queries, dashboards	dbt models on BigQuery, Airflow DAGs, SQL-based analytics pipelines; fraud scoring metrics in Spanner
SQL + Python	Python throughout the stack; SQL in rules monorepo (`get_shop_parameters.sql`), dbt models, central SQL definitions
2+ years relevant experience	3+ years on Yofi/Botnot antifraud platform (Feb 2022 — Oct 2025) + 15 years financial risk and controls
Understanding of payment processes and fraud typologies	Treasury, banking operations, payment execution, BIN/card validation — direct domain knowledge
Mathematical statistics & probability theory	PhD in Physics & Mathematics

CORE SKILLS

Antifraud & Detection Systems

Fraud rule design, YAML-based rule configuration and threshold tuning, allowlist/blocklist management, multi-signal order validation (BIN, IP, email, phone, user agent, address), bot/abuse/return fraud/resell fraud typologies, real-time severity scoring

ML-backed Fraud Scoring

SageMaker model routing, shadow prediction pipelines, `PredictionLevel` frameworks, feature engineering for fraud signals, customer and order abuse scoring (bot, return, resell, FTID, claim, fake profile)

Data & Analytics

Python (3.9–3.12), SQL, dbt (BigQuery), Airflow, Spark, Google Cloud Spanner, MongoDB, BigQuery; Advanced Excel, MS SQL for financial analytics

Payment & Banking Domain

BIN validation, payment execution, treasury operations, banking documentation, account management, card/transfer flows, cross-border transactions, financial compliance (IFRS/CAS)

Infrastructure & Tooling

AWS Lambda (SST/CDK/SAM), SQS/SNS/EventBridge, MongoDB, Google Cloud Spanner, Git/GitHub Actions, pre-commit hooks, pytest

WORK EXPERIENCE

Data Engineer / Antifraud Platform — Yofi Inc. (Botnot platform) (USA, remote)

February 2022 — October 2025

Product: Anti-fraud / customer intelligence platform for **Shopify** merchants (enterprise customers including **Lululemon**). Core mission: detect and prevent bot purchases, return fraud, discount abuse, reseller abuse, and fake profiles in real time.

Fraud Rules & Threshold Management

- Owned and maintained `yofi-rules-monorepo` — the single source of truth for all fraud model parameters across clients. Defined per-shop YAML rule configurations (`shopify\_return\_abusers\_rule`, `shopify\_return\_fraudsters\_model`, allowlisted domains, `min\_return\_rate` thresholds), CLI tooling to sync parameters between Python runtime and dbt SQL macros, and GitHub Actions for automated PR-based rule updates.
- Tuned fraud thresholds in production: added conservative shop lists, adjusted return parameters (`min\_return\_note\_rate`, `min\_fraud\_tags`), managed allowlisted email domains — directly balancing detection coverage against false positive rate.

ML Bot Detection & Abuse Scoring

- Contributed to `botnot-lambda-ml-bot-detection` — SageMaker-backed Lambda scoring each order for `is\_bot\_score`, `discount\_abuse`, refund probability. Implemented `is\_blacklisted`, `is\_whitelisted`, `is\_bad\_actor` decision logic; managed trust/risk signal system (`trust\_and\_risk\_messages`) driving downstream actions.
- Extended `yofi-lambda-ml-gateway` — ML routing Lambda selecting models per order/customer context: `ShopifyClaimAbusersModel`, `ShopifyResellersModel`, `ShopifyReturnAbusersModel`, `ShopifyReturnFraudstersModel`, LLM-based abuse scoring, FTID fraud scoring. Persisted shadow predictions to **MongoDB** for offline validation and back-testing.
- Contributed to `yofi-realtime-severity-engine` — real-time fraud severity classification feeding downstream alert workflows.

Order Validation Pipeline

- Maintained `botnot-lambda-order-validations` — multi-signal order risk assessment: IP geolocation validation, user agent analysis, credit card **BIN validation**, email validation, shipping/billing address checks, phone number verification. Integrated validation results into downstream fraud decision pipeline.

Feature Analytics & Back-testing

- Shipped new return analytics in `yofi-lambda-feature-analytics`: return rate calculations, refund line items analysis, fuzzy customer pattern detection, `ReturnAnalyticsProcessor` for new fraud typologies. Persisted analytics to **Google Cloud Spanner** for batch model consumption.
- Built dbt models in **BigQuery** via `yofi-dbt-models` and `yofi-rules-monorepo` dbt package — SQL-based rule logic and parameter macros used for back-testing rule efficiency across historical order data.

Data Platform

- Operated **Airflow** DAGs and **Spark** workloads on Kubernetes for fraud data pipelines (GCS, BigLake, Hudi/Delta table formats).
 - Extended **Airbyte** connectors for Shopify historical ingestion and partner data sources (Klaviyo, Moonsense).
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DataOps / AI Platform Engineer — Sinoptics AI (remote)

October 2025 - May 2026

- Built **AI-assisted document validation** workflows (n8n, Dify): OCR extraction, LLM structuring, compliance report generation including **risk assessment** and validation outputs.
 - Operated layered microservice architecture (orchestrator → agents → aggregator) with **RBAC**, audit logs, and regulatory compliance awareness (GDPR/CN).
 - Deployed FastAPI + Celery + Redis to K3S on AWS China, automated Yandex Cloud infrastructure (Cloud Functions, managed PostgreSQL, Keycloak).
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CFO — Shanghai Aircraft Design Limited Company (Shanghai, China)

November 2017 — 2022

- Designed multi-layer payment authorization controls, spending limits, and compliance frameworks from scratch for a new international entity.
- Full-cycle banking relationship management: account administration, payment execution, dispute resolution with financial institutions.

Executive Director — Engineering Solutions / New Technologies (China / Russia)

2008 — 2017

- Cross-border trade operations: counterparty due diligence, payment flow control, pricing and contract compliance across Russian and Chinese regulatory environments.
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EDUCATION

- **PhD in Physics and Mathematics**, Southern Federal University, 2000–2003

Probability theory, statistical modeling, mathematical analysis

- **Physics Diploma (Specialist)**, Southern Federal University, 1994–1999
 - **MBA**, RANEPA, 2005–2007
 - **College of Radio-Electronic Instrumentation**, 1991–1995
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CERTIFICATIONS

- **CAP** — Certified Accountant Practitioner
 - **CPA Russia**: financial accounting, IFRS reporting, management accounting
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